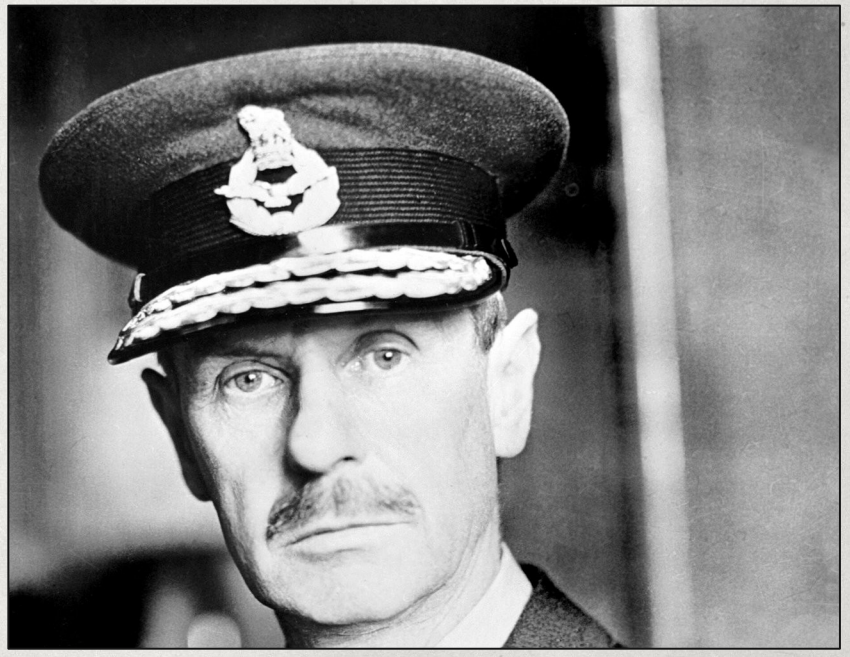
Tomorrow's Leaders

Special edition: effective communication — Hugh Dowding and the Battle of Britain, 1940

The battle won by information

How Hugh Dowding's air defence network helped a smaller air force hold off a larger one, and what a student team can learn from it

By Group [number]: [member names]



Air Chief Marshal Sir Hugh Dowding, Commander-in-Chief of RAF Fighter Command
Imperial War Museums, D 1417 (public domain)

Britain stands alone

Summer 1940: France has fallen, and the German air force is just across the Channel



Spitfires of No. 610 Squadron from Biggin Hill in "vic" formation, 24 July 1940
Imperial War Museums, CH 740 (public domain)

10 May Germany attacks France, Belgium and the Netherlands.

22 June France signs an armistice with Germany.

10 July The Luftwaffe, the German air force, begins its attack on Britain. Germany cannot invade until the Luftwaffe has defeated the RAF.

If you commanded Britain's fighters, where would you send them?

Tomorrow's Leaders

Part 1

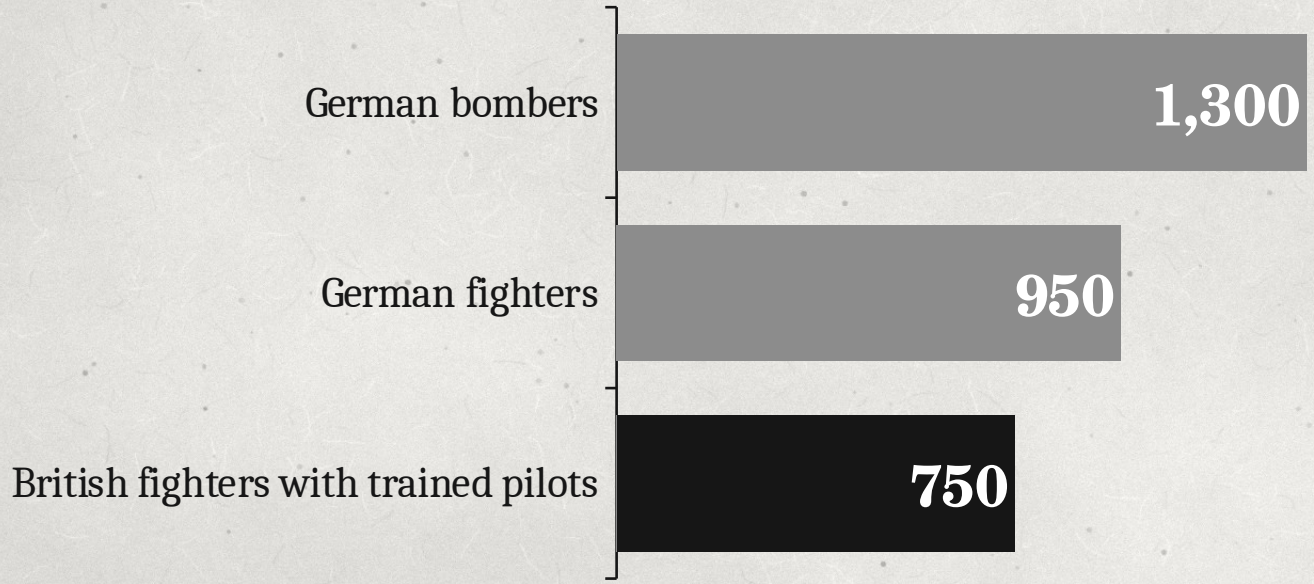
The story

How did Britain meet every raid in time, with no fighters to spare?

1

750 British fighters, 2,250 German aircraft

Strength on 10 August 1940, counting only aircraft ready to fly



The source gives German strength as "over" 1,300 bombers and "over" 950 fighters. Britain had 1,100 fighters ready to fly, but trained pilots for only about 750 of them.

Britain and Germany had similar numbers of fighters. But Britain's fighters also had to find and stop more than 1,300 bombers.

A squadron sent to the wrong place could not reach the real raid in time.

Gleave (1991, p. 48)

Twenty minutes to meet each raid

11 Group warned its controllers that raiders could be over London 20 minutes after the first radar plot

20 min

minutes of warning from a Kent radar station for a raid forming over France

4 min

minutes for the network to mark that raid on the operations room table

2 min

minutes that the 11 Group controller sometimes had to decide where to send fighters

Every minute spent passing a report along was a minute less for the fighters to reach the raid.

"The Battle of Britain" (2015); Battle of Britain Bunker (n.d.)

Seeing the raid was not enough

A radar operator sees 100 aircraft crossing the Channel. What happens next?

Option A

Phone the nearest airfield

A few fighters take off alone. Nobody else knows where the raid is going, and other airfields wait for orders.

Option B

Report to headquarters

Each level studies the report and passes an order down. The order reaches the pilots after the bombs have fallen.

The hard part was not seeing the raid. It was getting the right report to the right person in time.

Thought experiment based on the timings on slide 5.

Our group chats have the same problem

Amy posts in a five-person group chat: "Can someone do the theory part?" What happens next?

Option A

Nobody replies

Each member thinks someone else will do it. On Thursday, the theory part is still empty.

Option B

Two members both start

They write the same section twice, and nobody starts the comparison table.

Like the radar operator, Amy has the information. The hard part is getting it to the right person, in a form that person can act on.

A typical group project example.

The man who built the network

Air Chief Marshal Hugh Dowding, head of RAF Fighter Command from 1936 to 1940



Dowding (centre, hands behind his back) as King George VI decorates a pilot at Hornchurch, 27 June 1940
Imperial War Museums, CH 442 (public domain)

1935

As the RAF's head of research, he backs radar after a successful test near Daventry.

1936

He becomes the first head of Fighter Command, at Bentley Priory near London.

1936 to 1940

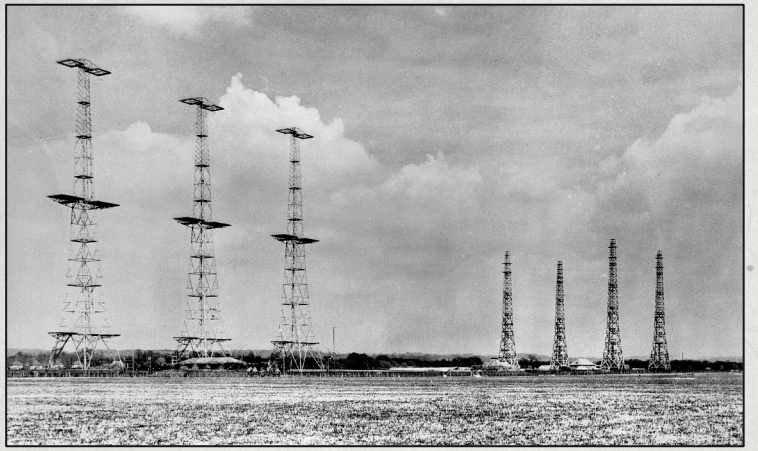
He links radar stations, observers, operations rooms and pilots into one network.

His idea, in our words: send each squadron to a raid that someone has already reported, instead of sending it to search the sky.

Bournemouth University (n.d.); Royal Air Force Museum (n.d.-a); Imperial War Museums (n.d.-b)

Step 1: Report

Radar stations covered the sea, volunteers covered the land, and both reported into one network



Chain Home radar station at Poling, Sussex, photographed in 1945
Imperial War Museums, CH 15173 (public domain)



An observer reports by chest telephone at Kings Langley, 1941 to 1945
Imperial War Museums, CH 8215 (public domain)

Chain Home radar

About 20 stations on the coast. They could not track aircraft over land, and the aircraft positions they reported, called plots, were not always accurate.

Observer Corps

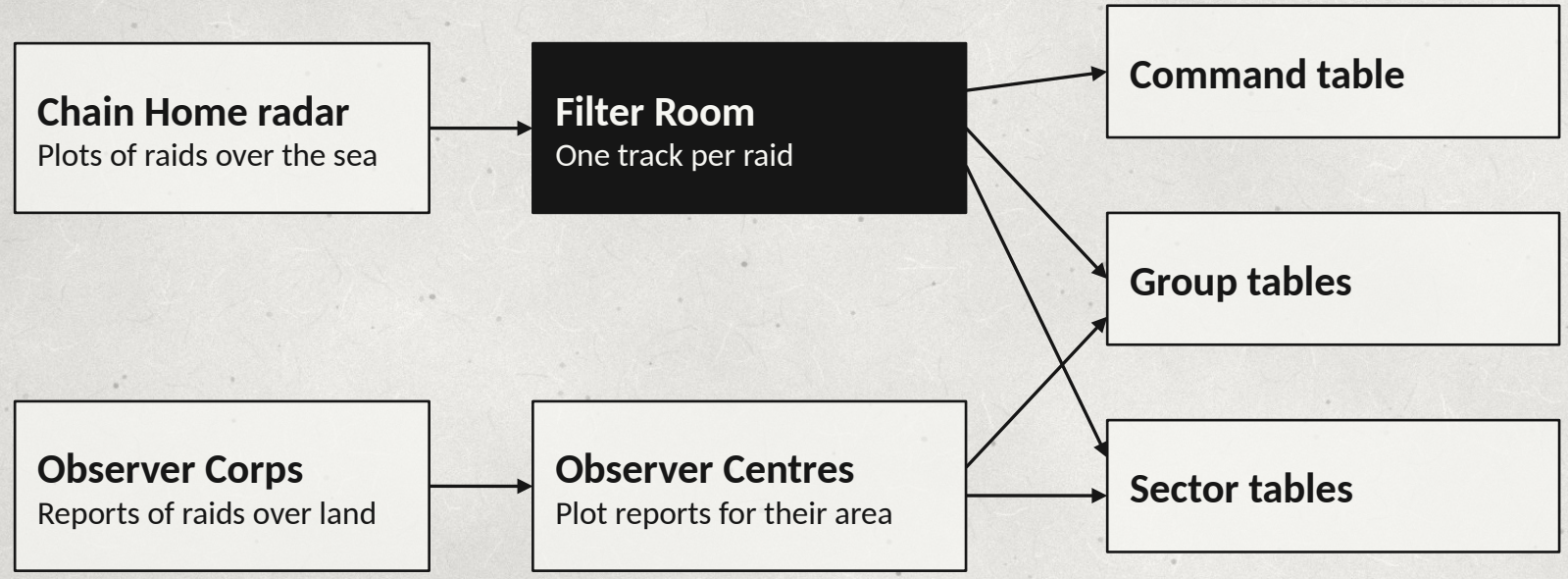
About 30,000 volunteers at about 1,000 posts. They spotted aircraft by eye and reported them by telephone.

Next problem Several radar plots often showed the same raid, and some gave it a different position or strength.

Historic England (2020); Royal Air Force Museum (n.d.-b); Imperial War Museums (n.d.-a)

Step 2: Filter

In the Filter Room, staff compared the radar plots and kept one track per raid



At the same time

Each filtered track went by telephone to the Command, Group and Sector tables at the same moment, so every level saw the same raid.

Next problem Commanders at three levels had to read their tables quickly and know how old each plot was.

Dowding (1946); Imperial War Museums (n.d.-b)

Step 3: Display

Every operations table used the same symbols



The operations room at Fighter Command headquarters, Bentley Priory (date not recorded)
Imperial War Museums, C 1869 (public domain)



Sector clock

Its red, yellow and blue segments each span 5 minutes. Each marker on the table took the colour of the segment in which it was placed, so a controller could see how old each plot was.



Raid block

A block on the table showed the raid's name and estimated strength, and an arrow showed its direction.



Tote board

A board showed the state of each squadron, such as “at readiness” or “left ground”.

Next problem A pilot in a cockpit needed his orders in a few words.

Royal Air Force Museum (n.d.-c); Imperial War Museums (2020, n.d.-b)

Step 4: Order

A sector controller gave each squadron its course and height by radio, in a few code words



Pilots of No. 19 Squadron stage a "scramble" for the camera at Fowlmere, September 1940
Imperial War Museums, CH 1398 (public domain)

Code word	Meaning
Scramble	Take off
Vector 230	Fly a course of 230 degrees
Angels 20	Fly at 20,000 feet
Pancake	Land

Group headquarters decided which sector sent fighters to each raid, and how many. The sector controller directed the pilots.

Next question Did the network work?

Dowding (1946)

"We had nothing like it"

Adolf Galland, a German fighter commander in 1940, on the British network

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"From the first the British had an extraordinary advantage, never to be balanced out at any time during the whole war, which was their radar and fighter control network and organisation. It was for us a very bitter surprise. We had nothing like it."

Galland (as cited in Wood, 1991)

Almost every squadron found its raid

15 September 1940: two large German daylight attacks on London, and both were beaten off

50 of 51 squadrons scrambled that day made contact with the raiders

56 vs 28 aircraft lost, German against British. 12 RAF pilots were killed.

17 Sept Hitler postponed the invasion of Britain until further notice



The preserved No. 11 Group operations room at Uxbridge, photographed in 2010
Photo: Danielstirland, CC0, via Wikimedia Commons

Churchill watched this table that day. "I now asked, 'What other reserves have we?' 'There are none,' said Air Vice Marshal Park."

Price (2003); Churchill (as cited in Glines, 1990)

Britain did not have more. It wasted less.

It wasted less because the right information reached the right person in time

Right information	The Filter Room kept one track per raid and removed duplicate plots.
Right person	Each track reached the Command, Group and Sector tables, and Group chose which sector met the raid.
In time	A raid appeared on the operations table about 4 minutes after radar detected it.
In a form they can act on	"Vector 230, Angels 20" told a pilot where to fly, in a few words.

Our definition Effective communication means the right information reaches the right person in time, in a form they can act on.

Where it comes from Keyton (2011): communication transmits "information and common understanding from one person to another". Spitzberg (2013): communication is competent when the other person judges it effective and appropriate. Neither mentions time, so we added it from the story.

Keyton (2011, as cited in Lunenburg, 2010); Spitzberg (2013); Dowding (1946); "The Battle of Britain" (2015)

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Part 2

Two theories

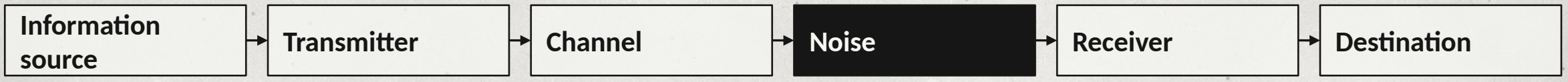
Why did the network waste so little, and why does Amy's chat waste so much?

2

Theory 1: Shannon and Weaver

A message travels from a source to a destination, and noise can distort it on the way

The model



In Amy's chat

Information source	Transmitter	Channel	Noise	Receiver	Destination
Amy	Amy types her request	The group chat	Dozens of other messages	Five phones	The member who should act, but nobody knows who

Weaver's three levels of communication problems

A. Technical

Can the symbols be sent accurately?

Amy's message arrives, but under dozens of others.

B. Semantic

Do the symbols carry the intended meaning?

"The theory part" means different things to different members.

C. Effectiveness

Does the meaning change behaviour as intended?

Nobody starts the work.

Shannon & Weaver (1949)

Theory 2: The 7 Cs

Shannon and Weaver's model follows the path of a message. The 7 Cs are questions about the message itself.

Clarity

What is the one main point?

Correctness

Are the facts and tone right?

Conciseness

Can it be said in fewer words?

Concreteness

Is it backed by specific facts?

Completeness

Does the reader have all they need to act?

Consideration

Is it written from the reader's side?

Courtesy

Is it respectful, even when firm?

Murphy, Hildebrandt and Thomas (1997) list seven qualities of a good message. We phrase each as a question.

Murphy et al. (1997)

Tomorrow's Leaders

Part 3

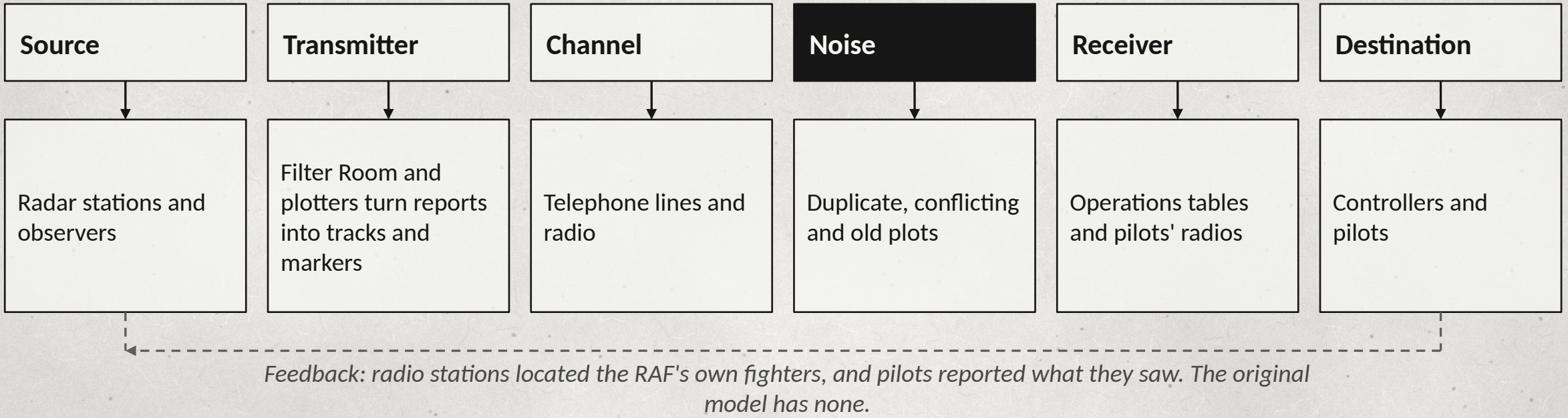
The theories applied

Does Dowding's network pass both theories, and does Amy's chat?

3

The Filter Room removed the noise first

Each part of Shannon and Weaver's model matches a part of the network



Feedback: radio stations located the RAF's own fighters, and pilots reported what they saw. The original model has none.

The difference from Amy's chat In the network, the Filter Room removed duplicate plots, and Group chose which sector acted. In Amy's chat, every message reaches all five members unfiltered, and nobody knows who should act.

Shannon & Weaver (1949); Dowding (1946); Imperial War Museums (n.d.-b)

The radio order passes; Amy's request fails

"Scramble. Vector 230. Angels 20." against "Can someone do the theory part?"

	Radio order	Amy's request	
Clarity	●●●	●○○	*Amy names no one*
Correctness	●●●	●●●	
Conciseness	●●●	●●●	
Concreteness	●●●	●○○	*No deadline or detail*
Completeness	●●●	●○○	*Nothing to act on*
Consideration	●●●	●○○	*The reader must guess*
Courtesy	●○○	●●●	*Amy is polite*

19 vs 13

The order scores higher on every C except courtesy. Amy's request scores higher only on courtesy.

In the 7 Cs, every quality has equal weight, but the right weight depends on the situation. In a cockpit, speed mattered more than courtesy. In a group chat, courtesy matters.

Dowding (1946); Murphy et al. (1997). Scores are our group's judgment.

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Part 4

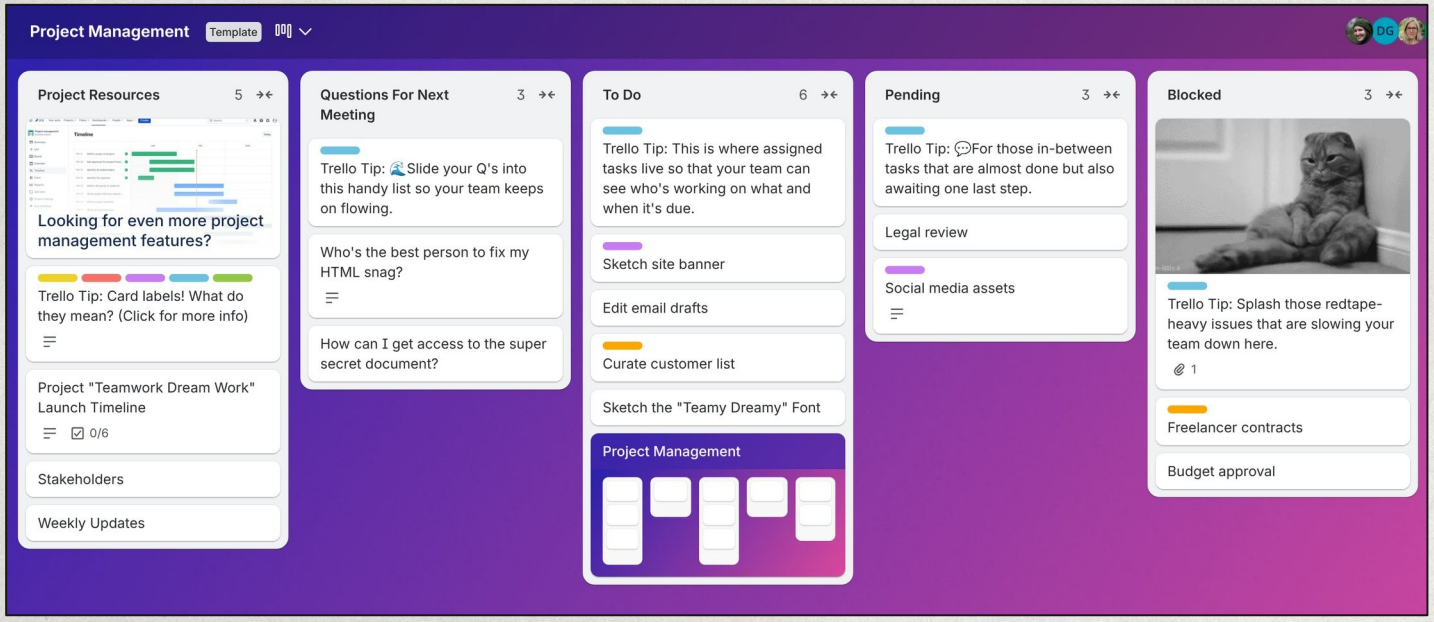
Lessons for us

4

Amy's team cannot get more members or more time.
How can it waste less?

Look familiar? It is a task board

Fighter Command's operations table worked like the project boards we use today



Trello's Project Management template board, captured 11 November 2026
Screenshot: trello.com/b/1x4Uql2u

1940	Today
One table for every level	**One board for the team**
Raid block	**Card**
Tote board	**Lists: To Do, Pending, Done**
The 20-minute limit	**Due date on the card**
Order to one squadron	**Card assigned to one member**

Notion, Microsoft Planner and GitHub Projects work the same way.

Royal Air Force Museum (n.d.-c); Imperial War Museums (2020); Trello (n.d.)

Practice 1: One task board for the project

Set it up like the operations table: one board, one owner per card, the same few lists, every deadline in view

1. **One board for the whole project**
   Every task is a card. If a task is not on the board, nobody will work on it.

2. **One owner and one due date per card**
   "Theory slides, Ming, Thursday."

3. **Four lists**
   Not started, Drafting, Ready for review, Done.

4. **A roadmap of deadlines**
   Every deadline on one timeline, so the team sees the time left.

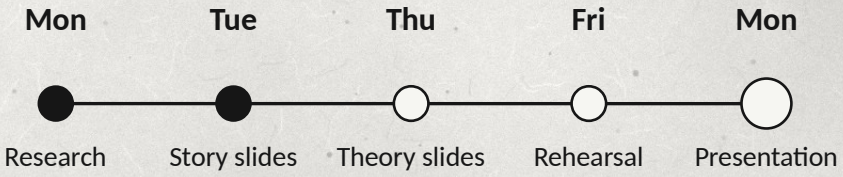
1940: one table for every level

1940: the raid block

1940: the tote board

1940: the 20-minute limit

Amy's roadmap



Filled: done. Hollow: still to come.

Evidence In a flight simulation study, pairs who shared an understanding of the task and of each other worked together better and performed better.

Where it works Any team that shares work can use a free tool such as Trello, Notion or Microsoft Planner.

Mathieu et al. (2000)

Practice 2: Pass updates through the board

Follow the network's path: report once, filter in one place, then send each request to one person

1. **Report once**
   Update the card. Do not repeat it in the group chat.

   1940: one network

2. **Filter in one place**
   Twice a week, one member, the coordinator, merges duplicate cards and closes old ones.

   1940: the Filter Room

3. **Assign, do not ask**
   Mention one person on the card: “@Ming, theory slides by Thursday 8 pm, please.”

   1940: an order to one squadron

4. **Confirm**
   The owner replies “Got it” or moves the card to Drafting.

   1940: pilots reported back

In the group chat

> *Amy: Can someone do the theory part?*

Five people read it. Nobody acts.

On the board

Theory slides
@Ming · Thu 8 pm · Drafting

One person reads it. One person acts.

Evidence Simpler letters to 35,050 tax filers led more of them to claim a benefit they were eligible for. Listeners who paraphrased made speakers feel more understood.

In Hong Kong Hall (1976) describes cultures like Hong Kong's as high-context: much of a request is left unsaid. A card that names one person can feel too direct, so add a “please”.

Bhargava & Manoli (2015); Weger et al. (2014); Hall (1976); Cardon (2008)

Tomorrow's Leaders

Germany had more aircraft. Britain wasted less.

Your team cannot get more members or more time. It can waste less by getting the right information to the right person in time.

- One board, read by everyone
- Update the card, not the chat
- Assign each task to one person

Questions for you

Where does information get lost in your group projects?

Does your team have a board, or only a group chat?

References

Battle of Britain Bunker. (n.d.). *Virtual bunker*. Retrieved November 11, 2026, from https://battleofbritainbunker.co.uk/virtual-resources/

The Battle of Britain: A not so narrow margin. (2015). *Air Power Review, 18*(2). https://www.raf.mod.uk/what-we-do/centre-for-air-and-space-power-studies/aspr/apr-vol18-iss2-7-pdf/

Bhargava, S., & Manoli, D. (2015). Psychological frictions and the incomplete take-up of social benefits: Evidence from an IRS field experiment. *American Economic Review, 105*(11), 3489–3529. https://doi.org/10.1257/aer.20121493

Bournemouth University. (n.d.). *The Daventry demonstration*. Retrieved November 11, 2026, from https://histru.bournemouth.ac.uk/Oral_History/Talking_About_Technology/radar_research/the_daventry_demonstration.html

Bridge, P/O. (n.d.). *The Royal Observer Corps, 1939-1945* [Photograph]. Imperial War Museums (CH 8215). Wikimedia Commons. https://commons.wikimedia.org/wiki/File:The_Royal_Observer_Corps,_1939-1945._CH8215.jpg

Cardon, P. W. (2008). A critique of Hall's contexting model: A meta-analysis of literature on intercultural business and technical communication. *Journal of Business and Technical Communication, 22*(4), 399–428. https://doi.org/10.1177/1050651908320361

Danielstirland. (2010). *Plotting table* [Photograph]. Wikimedia Commons. https://commons.wikimedia.org/wiki/File:Plotting_Table.jpg

Daventry, B. J. (1940). *Supermarine Spitfire Mark IAs of No 610 Squadron, Royal Air Force based at Biggin Hill, Kent, flying in three 'vic' formations* [Photograph]. Imperial War Museums (CH 740). Wikimedia Commons. https://commons.wikimedia.org/wiki/File:Supermarine_Spitfire_Mark_Is_of_No._610_Squadron_based_at_Biggin_Hill,_flying_in_%27vic%27_formation,_24_July_1940._CH740.jpg

Devon, S. A. (1940). *Pilots of No 19 Squadron, Royal Air Force 'scramble' for the photographer, from the back of a lorry at Fowlmere* [Photograph]. Imperial War Museums (CH 1398). Wikimedia Commons. https://commons.wikimedia.org/wiki/File:Pilots_of_No._19_Squadron_RAF_stage_a_mock_%27scramble%27_from_the_back_of_a_truck_at_Fowlmere_near_Duxford,_September_1940._CH1398.jpg

Dowding, H. C. T. (1946, September 10). The Battle of Britain [Despatch]. *Supplement to The London Gazette*, (37719), 4543–4571. https://www.thegazette.co.uk/London/issue/37719/supplement/4543

References (continued)

Friends of the Battle of Britain Monument. (n.d.). *Key facts about the Battle of Britain*. Retrieved November 11, 2026, from https://bbm.org.uk/the-battle/battle-key-facts/

Gleave, T. (1991). August and September 1940. In H. Probert & S. Cox (Eds.), *The battle re-thought: A symposium on the Battle of Britain*. Airlife. https://www.rafmuseum.org.uk/documents/Research/RAF-Historical-Society-Journals/Bracknell-No-1-Battle-of-Britain.pdf

Glines, C. V. (1990, September). Their finest hour. *Air Force Magazine*. https://www.airandspaceforces.com/article/0990finest/

Hall, E. T. (1976). *Beyond culture*. Anchor Press/Doubleday.

Historic England. (2020). *Chain Home transmitter tower, former RAF Stenigot* (List entry 1259301). https://historicengland.org.uk/listing/the-list/list-entry/1259301

Imperial War Museums. (2020). *Large print guide: Battle of Britain, Ops Block*. https://www.iwm.org.uk/sites/default/files/files/2020-09/OPS_BLOCK_Largeprint.pdf

Imperial War Museums. (n.d.-a). *How the Observer Corps helped win the Battle of Britain*. Retrieved November 11, 2026, from https://www.iwm.org.uk/history/battle-of-britain-observer-corps

Imperial War Museums. (n.d.-b). *What was the Dowding System?* Retrieved November 11, 2026, from https://www.iwm.org.uk/history/what-was-the-dowding-system

Lunenburg, F. C. (2010). Communication: The process, barriers, and improving effectiveness. *Schooling, 1*(1), 1–10.

Mathieu, J. E., Heffner, T. S., Goodwin, G. F., Salas, E., & Cannon-Bowers, J. A. (2000). The influence of shared mental models on team process and performance. *Journal of Applied Psychology, 85*(2), 273–283. https://doi.org/10.1037/0021-9010.85.2.273

References (continued)

Ministry of Information. (n.d.). *Air Chief Marshal Sir Hugh C T Dowding, Commander in Chief of RAF Fighter Command during the Battle of Britain* [Photograph]. Imperial War Museums (D 1417). Wikimedia Commons. https://commons.wikimedia.org/wiki/File:Hugh_Dowding.jpg

Murphy, H. A., Hildebrandt, H. W., & Thomas, J. P. (1997). *Effective business communications* (7th ed.). Irwin/McGraw-Hill.

Price, A. (2003). Battle of Britain Day. *Royal Air Force Historical Society Journal*, (29). https://www.rafmuseum.org.uk/documents/Research/RAF-Historical-Society-Journals/Journal-29-Dr-Alfred-Price-on-Battle-of-Britain-Day.pdf

Royal Air Force Museum. (n.d.-a). *Dowding: Architect of victory* [Podcast transcript]. Retrieved November 11, 2026, from https://www.rafmuseum.org.uk/documents/podcasts/transcripts/Dowding-Architect-Of-Victory.pdf

Royal Air Force Museum. (n.d.-b). *RADAR: The battle winner?* Retrieved November 11, 2026, from https://www.rafmuseum.org.uk/research/online-exhibitions/history-of-the-battle-of-britain/radar-the-battle-winner/

Royal Air Force Museum. (n.d.-c). *Sector room clock (replica)* [Museum object 71142]. Retrieved November 11, 2026, from https://collections.rafmuseum.org.uk/collection/object/object-71142/

Royal Air Force official photographer. (1940). *Flying Officer John Allen of No 54 Squadron receives the DFC from the King at Hornchurch* [Photograph]. Imperial War Museums (CH 442). Wikimedia Commons. https://commons.wikimedia.org/wiki/File:Royal_Air_Force_1939-1945-_Fighter_Command_CH442.jpg

Royal Air Force official photographer. (1945). *Chain Home radar installation at Poling, Sussex* [Photograph]. Imperial War Museums (CH 15173). Wikimedia Commons. https://commons.wikimedia.org/wiki/File:Chain_Home_radar_installation_at_Poling,_Sussex,_1945._CH15173.jpg

Royal Air Force official photographer. (n.d.). *The Operations Room at Headquarters, Royal Air Force Fighter Command, at Bentley Priory* [Photograph]. Imperial War Museums (C 1869). Wikimedia Commons. https://commons.wikimedia.org/wiki/File:The_Home_Front_in_Britain_during_the_Second_World_War_C1869.jpg

Shannon, C. E., & Weaver, W. (1949). *The mathematical theory of communication*. University of Illinois Press.

References (continued)

Spitzberg, B. H. (2013). (Re)Introducing communication competence to the health professions. *Journal of Public Health Research, 2*(3), Article e23. https://doi.org/10.4081/jphr.2013.e23

Trello. (n.d.). *Project management* [Board template]. Retrieved November 11, 2026, from https://trello.com/b/1x4Uql2u

Weger, H., Jr., Castle Bell, G., Minei, E. M., & Robinson, M. C. (2014). The relative effectiveness of active listening in initial interactions. *International Journal of Listening, 28*(1), 13–31. https://doi.org/10.1080/10904018.2013.813234

Wood, D. (1991). The Dowding system. In H. Probert & S. Cox (Eds.), *The battle re-thought: A symposium on the Battle of Britain*. Airlife. https://www.rafmuseum.org.uk/documents/Research/RAF-Historical-Society-Journals/Bracknell-No-1-Battle-of-Britain.pdf